

Midterm Project — V&V Addendum

Module M10 | Chapter 12 | Applies to midterm project submission

Module: M10 Chapter: 12 Weight: graded separately Due: End of Module M10

Overview

This addendum defines the required verification and validation evidence for the midterm project. Add a section titled **Verification and Validation** to your project memo and submit this addendum as a separate PDF (1–2 pages).

A model can only be considered valid relative to a stated purpose. This addendum forces you to apply the V&V checklist *after* the model is built, which is when most modellers discover the gaps they missed earlier.

1 Required Evidence

1.1 1. Model Purpose and Scope (15 pts)

State:

- The decision your model supports.
- The output measures used for that decision.
- The main exclusions from scope and the justification for each.

1.2 2. Verification Matrix (30 pts)

Include a table with at least **four** verification checks. At least one check must be **automated in code** (a `pytest` test or a notebook cell that can be rerun from a clean kernel).

Check type	What you tested	Expected	Observed	Pass?
Degenerate case	e.g., zero arrivals	No completions; zero queueing delay		
Extreme condition	e.g., very high capacity	Waiting time approaches zero		
Conservation check	e.g., arrivals = completions + WIP + departures	Counts balance at end of run		
Analytical benchmark or trace	e.g., M/M/1 formula or hand trace	Simulation matches within tolerance		

Fill in the “Expected” and “Observed” columns with the actual values from your model; do not leave them blank.

1.3 3. Trace-Based Debugging (20 pts)

Provide a short trace for a small fixed-seed or fixed-input run. The trace must be small enough to inspect by hand and must include:

- simulation time,
- entity identifier,
- event name,
- resource requested or released,
- queue length before or after the event, and
- sampled service time or routing decision when relevant.

Conclude with one sentence explaining *what the trace helped you verify*.

1.4 4. Validation Evidence (25 pts)

Use at least **two** of the following validation arguments. For each, state the evidence *and* the remaining limitation.

1. Face-validity review by a domain-aware person (e.g., classmate, instructor, or someone familiar with the system).
2. Comparison to historical or published data.
3. Sensitivity analysis showing that qualitative conclusions are stable across plausible parameter ranges.

4. Comparison to an analytical approximation (e.g., M/M/c formula).
5. Review of assumptions and exclusions with the decision maker.

1.5 5. Credibility Statement (10 pts)

End the V&V section with a brief credibility statement addressing:

- What decision is the model credible for?
- What decision should it *not* be used for?
- What additional data or review would most improve credibility?

Format

- Length: 1–2 pages (the table may extend this slightly).
- Submit as a separate PDF appended to or alongside the midterm report.
- File name: `midterm_vv_yournetid.pdf`

Grading Summary

Criterion	Points
Purpose, scope, and assumptions are explicit	15
Verification matrix: meaningful pass/fail evidence	30
Trace or automated test is reproducible	20
Validation evidence tied to the real decision context	25
Limitations and credibility statement are specific	10
Total	100

Full rubric: [rubrics/midterm_rubric.pdf](#) | Exemplar V&V addenda: the instructor package (available to instructors on request).